

Pickleball Court & Queue Management Engine: Comprehensive Technical Specification

Document Version: 2.0.0 **Status:** Revised — Incorporates On-Deck Shadow Courts, Player Editing, and Design System **Supersedes:** v1.0.0 **Target Platform:** Next.js (App Router), Supabase (PostgreSQL 16 + Realtime), Vercel Edge Network

Changelog from v1.0.0

Area	Change
Matchmaking	Added on-deck shadow courts — pre-composed matchups staged before a physical court is available, capped relative to active court count.
Admin UI	Court and on-deck matchups now render as individually selectable player cards (2x2 team-vs-team grid), not single-line pair strings. Players can be edited or replaced in place.
Data integrity	Player edits and replacements (name, rating, substitution) are explicitly unlogged by design — this is a deliberate product decision, not an oversight.
Queue model	Queue (all off-court checked-in players) and on-deck (host-visible pre-built matchups) are now distinct concepts. Queue feeds on-deck; on-deck feeds courts.
Design system	Added a full color palette specification (60-30-10 warm palette for admin/mobile, near-black high-contrast variant for the kiosk) with semantic-state color rules.
Open items	Flagged unresolved edge cases (singles on-deck layout, locked-pair edit permissions, reroll contention, host auth, queue-composition deadlock) as required pre-build decisions rather than leaving them implicit.

1. Executive Summary & Core Requirements Matrix

This specification details the technical, mathematical, and architectural blueprints for the **Pickleball Court & Queue Management Engine**. The platform automates court dispatch, player queuing, and rating calibration across up to 6 physical courts. It provides distinct views for court-side kiosks (TVs), session hosts (tablets/desktops), and players (mobile browsers via ephemeral QR access).

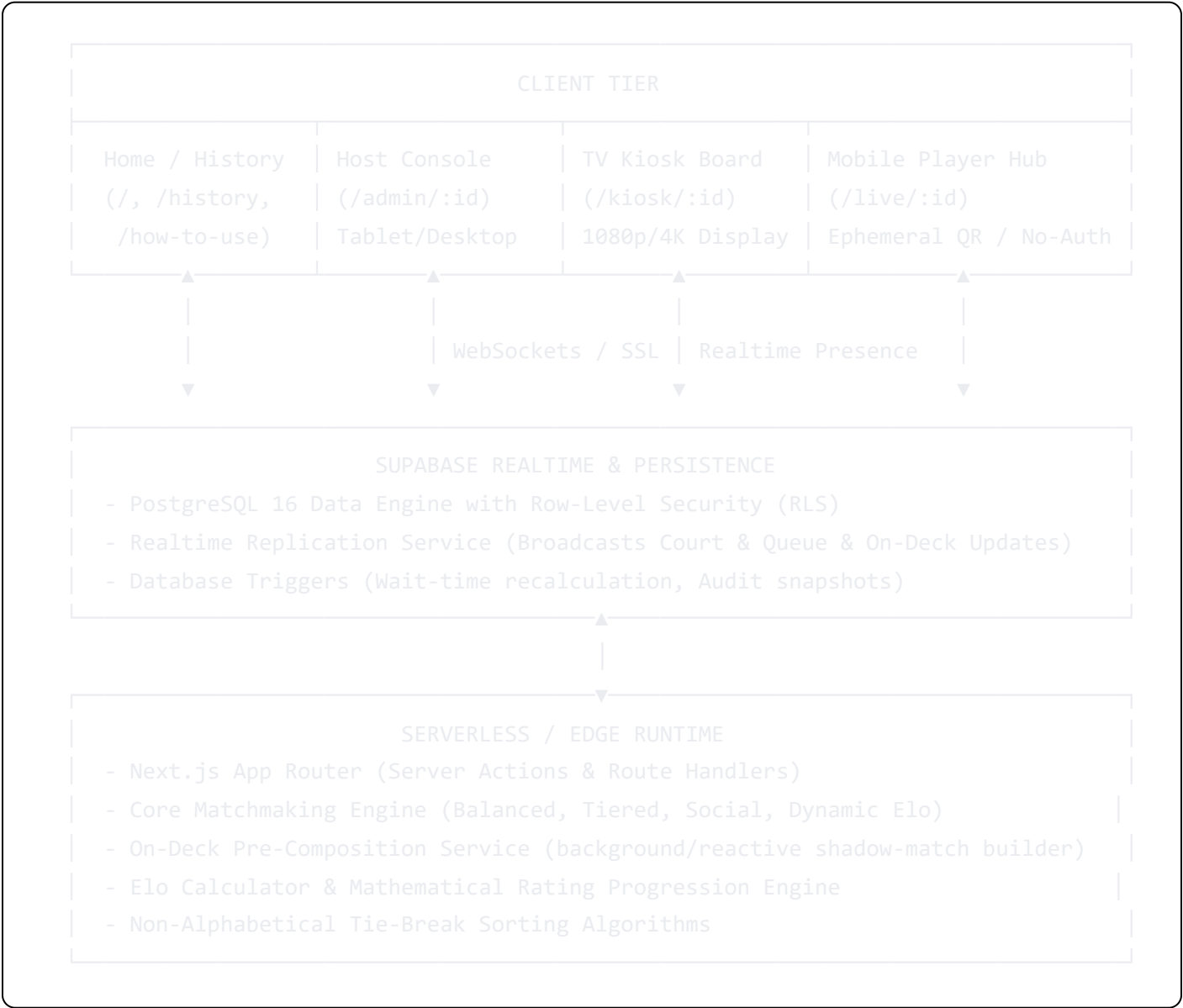
System Requirements Matrix

Requirement Area	Specification / Constraints
Court Capacity	Dynamic configuration from 1 to 6 concurrent physical courts.
Match Formats	Doubles (4 players), Singles (2 players), or Flexible per-court designation.
Rating System	Hybrid: Static Star Ratings (to in steps) alongside an automated dynamic Elo ladder.
Matchmaking Modes	4 selectable modes: Balanced, Skill-Separated, Social Mixer, and Dynamic Elo-Rated.
Pair Locking	Indivisible 2-player atomic grouping; calculates composite ratings and enforces pairing integrity.
On-Deck Staging	Host-visible, engine-composed shadow matchups (full Team A vs Team B), capped at <code>(active_courts - 1)</code> by default, called to a court manually — never auto-promoted.
Player Editing	Any player within an active court or on-deck matchup is individually selectable for name/rating edit or replacement, without audit logging.
Scorekeeping	Admin/host restricted. Global session switch: Scoring Required vs. Scoring Disabled/Off.
Timing Engine	Active match stopwatch (<code>(started_at)</code> count-up) + 90-second Summoning grace countdown.
Ergonomics	Self-service Rest/Hold mode; wait-time tracking; starvation-prevention algorithms.
Display Architecture	High-contrast 16:9 TV Kiosk display (near-black base), host admin console (warm palette), and zero-install QR mobile web app.
Historical Data	Session completion snapshots, tie-broken leaderboards, court utilization audits, and CSV export.

2. System Architecture & Technology Stack

The platform employs a reactive, edge-distributed architecture designed for rapid state synchronization across heterogeneous client devices over commercial venue Wi-Fi and mobile

networks.



Core Technologies

- **Frontend Framework:** Next.js (App Router), React 19, TypeScript.
- **Styling & UI:** Tailwind CSS, Lucide React (icons).
- **Audio Layer:** HTML5 Web Audio API (synthesized summoning chimes).
- **Database & Realtime:** Supabase (PostgreSQL 16 with Realtime WebSockets engine).
- **State Management:** Zustand for client-side local UI optimization.
- **Deployment:** Vercel Edge Platform.

3. Database Schema & Data Modeling

The relational model enforces data integrity via foreign keys, enumeration types, and conditional check constraints.

sql

```

-- ENUMERATIONS
CREATE TYPE court_status AS ENUM ('available', 'summoning', 'in_match', 'maintenance');
CREATE TYPE match_type AS ENUM ('doubles', 'singles', 'flexible');
CREATE TYPE match_mode AS ENUM ('balanced', 'skill_separated', 'social', 'elo_rated');
CREATE TYPE player_status AS ENUM ('checked_in', 'queued', 'staged', 'summoned', 'on_court');
CREATE TYPE match_stage AS ENUM ('on_deck', 'summoning', 'in_match', 'result_pending', 'completed');

-- SESSIONS
CREATE TABLE sessions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    join_pin VARCHAR(6) UNIQUE NOT NULL,
    name VARCHAR(100) NOT NULL,
    host_token VARCHAR(64) UNIQUE NOT NULL, -- see Section 11: Host Auth
    scoring_required BOOLEAN DEFAULT TRUE,
    grace_period_seconds INT DEFAULT 90,
    match_mode match_mode DEFAULT 'balanced',
    on_deck_cap_override INT DEFAULT NULL, -- NULL = use computed default (active_courts
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    ended_at TIMESTAMPTZ DEFAULT NULL
);

-- COURTS (Constrained to a maximum of 6 courts per session)
CREATE TABLE courts (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID REFERENCES sessions(id) ON DELETE CASCADE,
    court_number INT CHECK (court_number BETWEEN 1 AND 6),
    name VARCHAR(50) NOT NULL,
    status court_status DEFAULT 'available',
    assigned_match_type match_type DEFAULT 'doubles',
    current_match_id UUID,
    UNIQUE(session_id, court_number)
);

-- PLAYERS
CREATE TABLE players (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID REFERENCES sessions(id) ON DELETE CASCADE,
    name VARCHAR(100) NOT NULL,
    static_rating NUMERIC(2, 1) CHECK (static_rating IN (1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0)),
    current_elo INT DEFAULT 1200,

```

```

status player_status DEFAULT 'queued',
staged_match_id UUID DEFAULT NULL REFERENCES matches(id) ON DELETE SET NULL, -- non-n
wait_started_at TIMESTAMPTZ DEFAULT NOW(),
total_matches_played INT DEFAULT 0,
total_wins INT DEFAULT 0,
total_losses INT DEFAULT 0,
point_differential INT DEFAULT 0,
created_at TIMESTAMPTZ DEFAULT NOW()
);

-- LOCKED PAIRS
CREATE TABLE locked_pairs (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID REFERENCES sessions(id) ON DELETE CASCADE,
    player_1_id UUID REFERENCES players(id) ON DELETE CASCADE,
    player_2_id UUID REFERENCES players(id) ON DELETE CASCADE,
    composite_static_rating NUMERIC(2, 1) NOT NULL,
    composite_elo INT NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    CONSTRAINT distinct_pair_players CHECK (player_1_id <> player_2_id),
    UNIQUE(session_id, player_1_id),
    UNIQUE(session_id, player_2_id)
);

-- MATCHES
-- A match row now represents a matchup through its full lifecycle, including the
-- pre-court "on_deck" stage. court_id is NULL while stage = 'on_deck'.
CREATE TABLE matches (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID REFERENCES sessions(id) ON DELETE CASCADE,
    court_id UUID REFERENCES courts(id) DEFAULT NULL, -- NULL while on_deck
    stage match_stage NOT NULL DEFAULT 'on_deck',
    on_deck_slot_number INT DEFAULT NULL CHECK (on_deck_slot_number BETWEEN 1 AND 5), --
    match_mode_used match_mode NOT NULL,
    match_type match_type NOT NULL,
    team_a_ids UUID[] NOT NULL,
    team_b_ids UUID[] NOT NULL,
    score_a INT DEFAULT NULL,
    score_b INT DEFAULT NULL,
    elo_delta_team_a INT DEFAULT NULL,
    elo_delta_team_b INT DEFAULT NULL,
    staged_at TIMESTAMPTZ DEFAULT NOW(), -- when the on-deck matchup was composed

```

```

    summoned_at TIMESTAMPTZ DEFAULT NULL, -- set when host calls it to a court
    started_at TIMESTAMPTZ DEFAULT NULL,
    completed_at TIMESTAMPTZ DEFAULT NULL,
    match_duration_seconds INT DEFAULT 0
);

-- INDEXES
CREATE INDEX idx_players_queue ON players(session_id, status, wait_started_at);
CREATE INDEX idx_matches_session_completed ON matches(session_id, completed_at);
CREATE INDEX idx_matches_on_deck ON matches(session_id, stage) WHERE stage = 'on_deck';

```

Schema notes

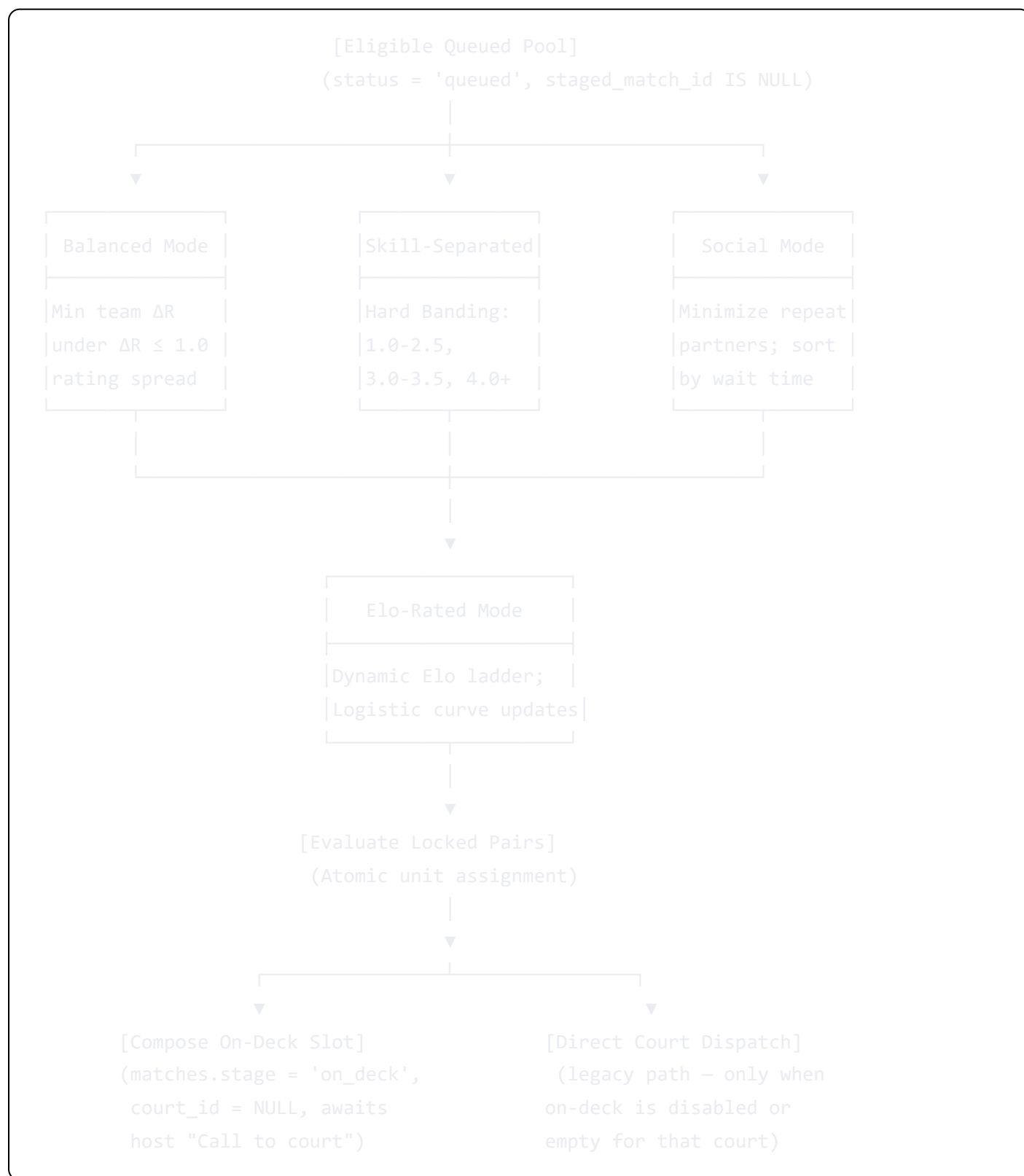
- **No substitution/edit log table exists by design.** Player name/rating edits and in-place replacements on active or on-deck matchups are direct `UPDATE` statements against `players` and `matches.team_a_ids` / `team_b_ids`. This is intentional — see Section 9 (Player Editing & Non-Logging Policy) — not an omission.
- `players.staged_match_id` prevents a player from being claimed by two on-deck slots simultaneously, and prevents the general dispatch engine from also drafting a staged player into a different matchup.
- `matches.stage` replaces the implicit state that was previously inferred from `court_id` and timestamp presence in v1.0.0. This makes the on-deck stage a first-class, queryable state rather than a derived one.

4. Algorithmic Matchmaking Specifications

The matchmaking engine now has **two triggers** instead of one:

1. **On-deck pre-composition** (new in v2.0.0): runs whenever the queue changes (check-in, rest-toggle, checkout) or an on-deck slot is emptied (called to court or manually cleared), attempting to fill available on-deck slots up to the cap.
2. **Direct dispatch** (legacy v1.0.0 path, retained for sessions/hosts that skip on-deck): still triggers when a court transitions to `available`, for the case where no on-deck matchup exists yet, or a host wants to bypass staging entirely.

Only players with `status = 'queued'` (i.e., not already `staged` into another on-deck slot) are eligible for either trigger.



1. Balanced Mode

- **Rating Envelope:** The individual static star spread among all 4 chosen players must not exceed 1.0 :

- **Objective Function:** Minimize side disparity:

Target condition: _____.

2. Skill-Separated Mode (Tiered Play)

- **Partitions:**
 - **Tier 1 (Recreational):** Static Star
 - **Tier 2 (Intermediate):** Static Star
 - **Tier 3 (Advanced):** Static Star
- **Court Selection Rule:** Players never cross tier boundaries. If multiple tiers have sufficient queued players, courts are assigned to the tier with the highest aggregate bench wait time.

3. Social Mode (Club Mixer)

- **Objective Function:** Maximize wait-time priority while penalizing repeat partner pairings:

Where _____ is the count of previous matches where Player _____ and Player _____ were on the same team during the active session.

4. Dynamic Elo-Rated Mode

- ****Initial Seed from Static Stars:****

(Yields: _____, _____, _____)

- **Team Composite Elo:**

- **Expected Outcome Formula:**

- **Rating Delta Formula:**

Where for a win and for a loss.

- for players with session matches (accelerated convergence).
- for established players (session matches).
- Both players on Team A receive ; both players on Team B receive (or the inverse if Team B wins).

5. Locked-Pair Calculus

- A locked pair forms an atomic unit occupying two roster slots in doubles.
- Combined metrics:

-
- **Imbalance Clamp:** If , the pair is barred from Balanced mode unless matched against an opposing locked pair with an equivalent composite rating (); otherwise, the pair is routed to Social or Elo-Rated mode.

6. On-Deck Slot Composition (New)

- **Slot Cap:** `on_deck_cap = session.on_deck_cap_override ?? (active_courts - 1)`, recomputed whenever `active_courts` changes. Example: 3 active courts → 2 on-deck slots.
- **Trigger conditions:** the on-deck composer attempts to fill any empty slot (up to the cap) whenever:
 - A player checks in, un-rests, or is newly eligible.
 - An on-deck slot is vacated (called to a court, or manually cleared by the host).
 - The host changes the active matchmaking mode.
- **Composition process:** identical mode logic to Sections 4.1-4.4, applied against the eligible pool, producing a full `matches` row with `stage = 'on_deck'`, `court_id = NULL`, and both `team_a_ids` / `team_b_ids` populated. Claimed players are marked `status = 'staged'` and `staged_match_id` is set, removing them from the pool for the next slot's composition **and** from the legacy direct-dispatch pool.
- **Slot independence:** slots are composed sequentially (Slot 1 before Slot 2) so that Slot 2's composition never contends with Slot 1 for the same players. A manual reroll of one slot (see below) releases its players back to the eligible pool before recomposing, but does not touch other slots' claimed players.
- **Manual reroll:** the host may reroll a single on-deck slot. This releases that slot's 2 or 4 players back to `status = 'queued'` and re-runs composition for that slot only, respecting other slots' existing claims.

- **Calling to court (host-initiated only):** the host selects an available court and one on-deck slot; on confirmation, the match row transitions `stage: 'on_deck' → 'summoning'`, `court_id` is set, `summoned_at = NOW()`, players transition `staged → summoned`, and the 90-second grace countdown begins. **On-deck matchups are never auto-promoted to a court** — this is a required manual action regardless of how long a court has been sitting available.
- **Stalled slot handling:** if fewer than the required number of eligible players remain to fill a slot (e.g., remaining queued players are locked pairs that cannot form a valid opposing side under the active mode), the slot renders as `stalled` rather than silently persisting at partial capacity — see Section 10, Open Item 5.

5. Leaderboard Engine & Non-Alphabetical Tie-Breaking

The leaderboard engine explicitly prevents alphabetical or random database sequential sorting when win rates or match counts are tied.

Mode Setting	Strict Hierarchical Tie-Breakers
Dynamic Elo Mode	1. Current Numerical Elo Rating 2. Total Net Point Differential 3. Total Match Volume 4. FIFO Check-In Timestamp
Standard Mode (Scoring: REQUIRED)	1. Win Percentage: $W / (W + L)$ 2. Point Differential (Net $\pm\Delta$) 3. Total Matches Played 4. Strength of Schedule (SOS) 5. FIFO Check-In Timestamp
Standard Mode (Scoring: DISABLED)	1. Total Completed Matches 2. Total Active Court Minutes 3. FIFO Check-In Timestamp

- **Strength of Schedule (SOS):** The average static star or Elo rating of all opponents faced during the session.
- **Locked-Pair Attribution:** Locked pairs appear as a unified entity during active partnership. Personal player records retain match volume, points, and individual Elo updates.

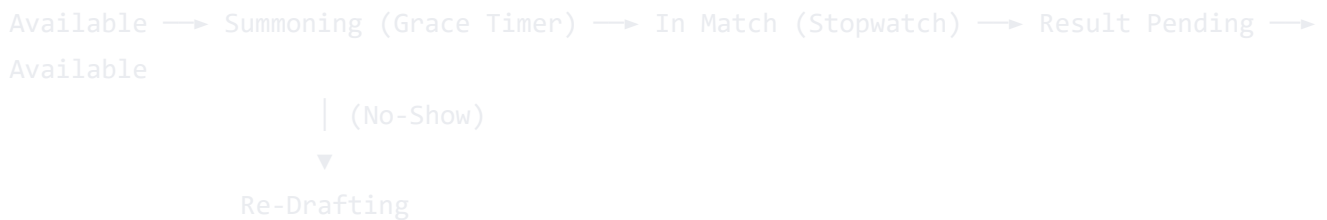
- **Silent-edit interaction:** if a player's name or rating is edited mid-session (Section 9), historical leaderboard rows reflect the *current* player record, not a point-in-time snapshot — since no edit log exists, there is no mechanism to reconstruct "rating at time of match" separately from "rating now." This is a direct consequence of the no-logging policy and should be communicated to hosts as expected behavior, not a bug.

6. Dual State Machine & Turnaround Lifecycle

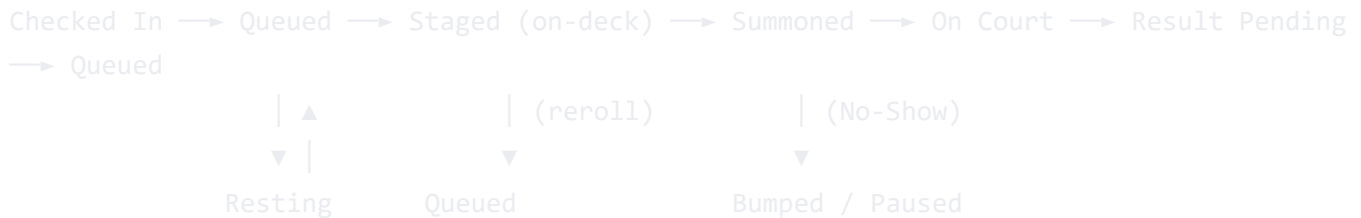
[MATCH STAGE STATE MACHINE – new in v2.0.0]



[COURT STATE MACHINE]



[PLAYER STATE MACHINE]



Turnaround Lifecycle Details

Step	State	Trigger	System Actions
0. On-Deck Composition	<code>match.stage = on_deck</code> <code>player.status = staged</code>	Queue change or slot vacancy triggers composer (Section 4.6).	Engine builds a full Team A vs Team B matchup; claims players via <code>staged_match_id</code> ; slot renders on admin console with individually selectable player cards.
1. Match Summon	<code>court.status = summoning</code> <code>match.stage = summoning</code> <code>player.status = summoned</code>	Host explicitly selects an on-deck slot and an available court, and confirms "Call to court".	Starts 90s grace countdown; emits TV audio alert; displays "Proceed to Court X" banner on mobile devices. Never happens automatically.
2. Match Start	<code>court.status = in_match</code> <code>player.status = on_court</code>	Host clicks "Start" OR grace timer hits <code>00:00</code> without flag.	Halts countdown; begins active match stopwatch (<code>started_at = NOW()</code>); UI switches to green/olive active status.
3. No-Show Flag	<code>player.status = resting</code> <code>court.status = summoning</code>	Host flags absent player during grace period.	Offending player moved to <code>resting</code> ; matching engine drafts the next eligible player in without dropping remaining players.
4. Completion (Scores ON)	<code>court.status = result_pending</code>	Match ends; host opens score prompt.	Match stopwatch stops; host inputs scores (e.g., 11-9); Elo and differentials update; court set to <code>available</code> .
5. Completion (Scores OFF)	<code>court.status = available</code>	Match ends; host clicks "Finish Match".	Match stopwatch stops; match count increments; player records update; court immediately set to <code>available</code> .
6. Rest Toggle	<code>player.status = resting</code>	Player or host toggles "Take a Break".	Player excluded from queue and on-deck composition algorithms; original check-in timestamp and match stats remain intact.
7. Player Edit (In Place)	No state change	Host taps a player card on an active court or on-deck	Direct update to <code>players</code> row (and <code>matches.team_a_ids</code> / <code>team_b_ids</code>) if

Step	State	Trigger	System Actions
		slot and edits name/rating, or replaces the player.	replaced). No audit record is created — see Section 9.

7. Application Page & Route Specifications

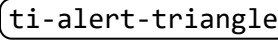
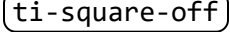
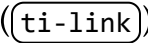
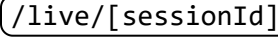
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/
├─ / (Home Portal)
├─ /how-to-use (System Operations & Rules Manual)
├─ /history (Archived Sessions Index)
├─ /history/[sessionId] (Session Audit, Leaderboard & CSV Export)
├─ /admin/[sessionId] (Host Console)
├─ /kiosk/[sessionId] (1080p/4K TV Display Board)
└─ /live/[sessionId] (Mobile Player Hub)
```

1. Home Page (/)

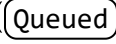
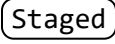
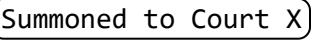
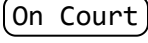
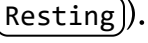
- **Host Setup Modal:**
 - Captures: Session Name, Court Count (1-6), Mode (**Balanced**, **Skill-Separated**, **Social**, **Elo-Rated**), and Scoring Toggle (**Required** vs. **Off**).
 - Yields new **session_id**, unique 6-character **join_pin**, and a private **host_token** (see Section 11).
- **Player Access:**
 - Alphanumeric PIN input or mobile QR scanner.
- **Auto-Resume Detection:**
 - Reads **localStorage** for active session IDs; renders direct chips: **[Resume Host Console]** or **[Open Player Queue]**.

2. Live TV Kiosk Board (/kiosk/[sessionId])

- **Aspect Ratio:** 16:9 widescreen layout for 1080p and 4K displays.
- **Color base:** Near-black (**#0D0D0C**), not a warm/umber tint — chosen specifically to avoid reading as a brown/muddy tone under venue lighting. Card panels sit one step up (**#161614**).
- **Upper Section (Courts 1-6 Grid):**

- **Active Courts:** Olive ( #8A9A5B) border,  icon, format badge, player names, star rating/Elo, and active match stopwatch (.
- **Summoning Courts:** Terracotta ( #C36F42) border,  icon, synchronized 90s countdown timer, and alert banner.
- **No-Show / Alert state:** Reserved semantic red ( #F09595), outside the core 4-color palette — deliberately breaks from the warm family since it must be visually distinct from "waiting") with  icon.
- **Inactive Courts:** Muted low-opacity cream on dark card,  icon, .
- Color is never the sole status signal — every state pairs a border color with a distinct icon for colorblind accessibility.
- **On-Deck visibility (decision required — see Section 10, Open Item 6):** whether upcoming on-deck matchups are shown read-only on the kiosk, or remain admin-only until called.
- **Lower Section (On-Deck / Queue):**
 - Horizontal queue cards displaying wait-time stopwatches, player star ratings, and locked-pair indicators (.
- **Venue Hub:**
 - Top-right QR code pointing to  for instant onboarding.

3. Mobile Player Hub (

- **Status Card:** Real-time state indicator (,  (on-deck), , , or .
- **Self-Service Controls:**
 - Single-tap "Take a Break / Resume Play" toggle.
 - "Request Pair Lock" selector (picks from checked-in players).
- **Metrics & Standings:**
 - Personal off-court wait timer.
 - Read-only live session leaderboard.
- **Color base:** Cream ( #F5F0E8) 60% / olive ( #8A9A5B) 30% surfaces / terracotta ( #C36F42) 10% accent for primary actions, umber ( #3E2F23) text throughout.

4. Host Admin Console (

Three distinct, vertically stacked zones, replacing the v1.0.0 "two-tap court cards + roster management" model:

- **Zone 1 — Courts:**

- Each active/in-match court renders players as **individually selectable cards in a 2x2 team-vs-team grid** (Team A column vs Team B column, "vs" divider), not a single-line pair string.
- Tapping a player card opens an action sheet: **Edit name**, **Edit rating**, **Replace player** (pulls from queue). Locked-pair players show a `(ti-link)` badge in place of the edit-menu icon — see Section 10, Open Item 2 for the unresolved edit-permission question.
- Available courts show a "Call match" action that is de-emphasized (visually muted) when on-deck slots exist, steering the host toward calling from on-deck rather than triggering legacy direct dispatch.
- Action triggers retained from v1.0.0: `(Start Match)`, `(No-Show / Re-Draft)`, `(Complete Match)`. Dynamic Score Modal unchanged (hidden when scoring is disabled).
- **Zone 2 — On-Deck Matchups:**
 - Displays up to `(on_deck_cap)` slots (header shows "`(N)` of `(on_deck_cap)` slots used" and the cap formula).
 - Each slot is a fully composed matchup rendered in the same 2x2 individually selectable player-card grid as Zone 1, dashed terracotta border to visually distinguish "not yet on a physical court."
 - Each slot has: a reroll action (`(ti-refresh)`, recomposes that slot only), and a primary "Call to court" action that prompts the host to pick a destination court.
 - A slot with insufficient eligible players to complete renders in a `(stalled)` visual state rather than persisting silently incomplete (see Section 10, Open Item 5).
- **Zone 3 — Queue:**
 - Full list of all off-court, checked-in players not currently claimed by an on-deck slot, each row showing wait time and rating.
 - This list is separate from Zone 2 — a player only leaves the Queue view when the on-deck composer claims them into a slot.
- **Roster Management:**
 - Add player with star rating picker (1.0–5.0).
 - Pair-locking builder (combines 2 players into an atomic unit).
 - Status overrides: Manually set any player to `(Resting)`, `(Queued)`, or `(Checked Out)`.
- **Session Settings Drawer:**
 - On-the-fly toggling of Matchmaking Modes and Scoring Requirements.
 - On-deck cap override (defaults to `(active_courts - 1)`; host may set explicitly).
- **Responsive behavior:** the 2x2 player-card grid (Zones 1 and 2) collapses to a single stacked column below tablet width, since four cards side-by-side do not fit a host's phone

screen.

5. How to Use Page (</how-to-use>)

- **Player Guide:** Check-in steps, QR scanning, grace periods, Rest mode etiquette, and locked pairs.
- **Host Manual:** Operating 6 courts efficiently, using on-deck staging, handling no-shows, odd player rotations, and configuring match modes.
- **Star & Elo Rating Scale Reference:**

Stars	Elo Range	Skill Benchmarks
1.0 - 2.0	600 - 950	Learning non-volley zone (kitchen) rules, scoring, and paddle control; frequent unforced errors.
2.5	950 - 1125	Can sustain short dink rallies; consistent serve direction; developing third-shot drop mechanics.
3.0	1125 - 1300	Consistent medium-pace dinking; understands kitchen positioning; struggles against high-speed attacks.
3.5	1300 - 1475	Directional control; aggressive dinking; consistent third-shot drops and resets under moderate pressure.
4.0	1475 - 1650	High-pace hand speed; anticipates court openings; low unforced error rate; advanced spin and resets.
4.5+	1650 - 2000	Elite positioning; exceptional soft touch; mastery of pace modulation, reset angles, and attack defense.

6. Session History Page (</history> & [/history/\[sessionId\]](/history/[sessionId]))

- **Session Archive (</history>):** Filterable list of completed sessions by date, court count, and match mode.
- **Audit Detail ([/history/\[sessionId\]](/history/[sessionId])):**
 - **Final Leaderboard:** Frozen table of final standings with all tie-breaker criteria, subject to the silent-edit caveat in Section 5.
 - **Court Utilization Audit:** Total active match duration vs. idle turnover time per court.
 - **Data Export:** Instant CSV export containing full match logs: [Match ID](#), [Court Number](#), [Mode](#), [Team A](#), [Team B](#), [Scores](#), [Elo Changes](#), and [Duration](#). Note:

exported Team A/Team B reflect final roster at completion time, not necessarily who started the match, due to the no-logging substitution policy.

8. Design System & Color Palette

Two palette contexts, sharing one 4-color warm base but applied at opposite contrast extremes for their respective viewing conditions.

8.1 Base palette

Color	Hex	Role
Cream	□ #F5F0E8	Dominant surface (60%)
Olive	■ #8A9A5B	Secondary surface / active-state (30%)
Terracotta	■ #C36F42	Accent / primary action / summoning-state (10%)
Umber	■ #3E2F23	Text color throughout (light-mode surfaces)

8.2 Admin console & mobile hub (light, warm)

- 60% cream background, 30% olive for active/secondary panels, 10% terracotta reserved for the single primary action per view (e.g., "Call match", "Call to court").
- Umber text at full opacity for primary labels, reduced opacity (~60–65%) for secondary/meta text (timestamps, helper copy).
- On-deck slots use a dashed terracotta border to signal "staged, not yet live" as a distinct visual category from solid-bordered active courts.

8.3 Kiosk board (dark, high-contrast)

- Base is **near-black** (■ #0D0D0C), explicitly not umber-tinted — an umber-derived dark background was tested and rejected for reading as a muddy/brown tone rather than a neutral dark surface.
- Card panels: ■ #161614. Nested elements (e.g., queue chips): ■ #0D0D0C with a low-opacity cream border.
- Cream text at full opacity for primary content, reduced opacity for secondary content — same pattern as the light mode, inverted.
- Status colors keep their hue (olive = active, terracotta = summoning) but are used as **borders and icon colors against the dark base**, not as fills, to preserve contrast at TV viewing distance.
- No-show/alert state uses a dedicated red (■ #F09595) outside the 4-color base palette — this is intentional, since it is the one state that must be unambiguous at a glance and the

base palette's warm hues sit too close together in luminance to safely encode "urgent problem."

8.4 Accessibility rule

Every status distinction (active / summoning / alert / closed) pairs a color with a distinct icon (`ti-player-play`, `ti-clock`, `ti-alert-triangle`, `ti-square-off` respectively). Color is never the sole differentiator, addressing the colorblind-accessibility gap identified during design review of the original all-color-coded kiosk concept.

9. Player Editing & Non-Logging Policy

This is a deliberate product decision, called out explicitly so it is not "fixed" by a future contributor assuming it was an oversight.

- **What's editable in place:** any player's name or star rating, on an active court or an on-deck slot, via tapping their individual card.
 - **What "replace" does:** swaps one player in a matchup for another player pulled from the queue, updating `matches.team_a_ids`/`team_b_ids` and both players' `status` fields directly.
 - **No audit trail is created for any of the above.** There is no `edit_log` or `substitution_log` table. The system does not attempt to reconstruct "who was originally scheduled" or "what rating a player had at match time" after an edit.
 - **Downstream consequence:** CSV exports and historical leaderboards (Section 5, Section 7.6) reflect final state only. Elo deltas already applied prior to an edit are not retroactively recalculated.
 - **Rationale (as discussed):** this keeps in-match corrections fast and frictionless for hosts running live sessions, at the deliberate cost of historical precision.
-

10. Open Items Requiring Resolution Before Build

These were surfaced during design review and are **not yet resolved** in this specification. Each should be explicitly decided and folded into a v2.1 revision before implementation begins.

1. **Host authentication model.** `sessions.host_token` is now included in the schema (Section 3) as a placeholder, but its issuance, storage (signed cookie vs. localStorage), and rotation policy are undefined. This is higher priority than in v1.0.0 given that player edits are now silent and unlogged — access control is the only remaining safeguard.
2. **Locked-pair edit permissions.** Can a host replace one half of a locked pair directly from a matchup card, or must the pair be unlocked first? Undefined.
3. **Singles on-deck layout.** The 2x2 team-vs-team grid assumes doubles. Singles on-deck slots (1v1) need their own card layout variant.

4. **Reroll contention.** Confirmed that slots compose sequentially and don't contend for players — but the UX of a reroll that empties a slot when the remaining queue can't refill it (see Item 5) is not yet designed.
 5. **Stalled on-deck slots / queue-composition deadlock.** If remaining queued players cannot form a valid matchup under the active mode (e.g., only mismatched locked pairs remain), a slot should render as `stalled` with an explanatory message rather than persisting incomplete or silently blocking. Exact messaging and host recovery actions (force-compose under relaxed constraints? route to Social mode?) are undefined.
 6. **Kiosk visibility of on-deck matchups.** Undecided whether players should see upcoming on-deck matchups on the TV board (reduces anxiety, but risks showing a matchup that changes before it's called) or whether on-deck stays admin-only until summoned.
 7. **Grace-period auto-no-show ambiguity (carried over from v1.0.0).** What happens if the grace timer expires with a genuinely absent player who was never manually flagged — does the match auto-start short-handed, or does the system need an explicit auto-no-show at `00:00`?
 8. **Session on-deck cap changes mid-session.** If a host reduces active court count mid-session such that existing on-deck slots exceed the new cap, is the excess slot cleared (players returned to queue) or grandfathered until called?
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11. Hardware, Network & Edge-Case Failure Recovery

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Failure Mode / Edge Case	Architectural Mitigation
-----+	
Player No-Show at Court Call	Host clicks "No-Show". Player moves to "Resting".
Matchmaking	
	engine selects next eligible candidate within
<500ms without	
	disrupting the remaining 3 players.
-----+	
-----+	
Odd Player Count Deadlocks	Maximum Consecutive Bench Rule: Any player who
sits out 1	
(e.g., 17 players on 4 courts)	full match cycle receives absolute queue priority
on the next	
	available court, bypassing mode constraints to
prevent stall.	
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Club Wi-Fi Interruption	Client maintains optimistic UI state. Timers
evaluate against	
	server timestamps (started_at). WebSockets auto-
reconnect via	
	exponential backoff with zero state loss.
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Rating Imbalance in Locked Pairs	Pairs with $ R1 - R2 \geq 1.5$ are restricted from
Balanced mode	
(e.g., 5.0 paired with 2.0)	unless an equivalent opposing pair is queued;
otherwise	
	automatically routed to Social or Elo-Rated mode.
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On-Deck Slot Stalls	See Section 10, Open Item 5. Slot renders
`stalled` rather	

| (no valid matchup composable) | than persisting incomplete; host recovery action
TBD. |

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| Unauthorized Admin Access | See Section 10, Open Item 1. Host token model
required before |

| (silent, unlogged edits at risk) | production launch – currently unresolved.

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12. Implementation Roadmap & Milestones

Phase 1: Foundations & Real-Time Sync (Weeks 1-2)

- └─ Supabase PostgreSQL schema, migrations, RLS policies, and triggers
- └─ Implement Elo rating math and initial star-to-Elo conversions
- └─ Resolve Open Item 1 (host auth model) before any admin-write endpoint ships
- └─ Build mock player generator for high-concurrency stress testing

Phase 2: Matchmaking Engine & Dual State Machines (Weeks 3-4)

- └─ Build TypeScript engine for Balanced, Tiered, Social, and Elo modes
- └─ Build On-Deck Pre-Composition Service (Section 4.6) as a distinct module
- └─ Implement locked-pair atomic handling and tie-breaking algorithms
- └─ Resolve Open Items 2, 3, 4, 5 (pair edits, singles layout, reroll, stalls)
- └─ Construct court/match/player state machines with on-deck stage, 90s grace countdowns, and stopwatches

Phase 3: Route & Interface Implementation (Weeks 5-6)

- └─ Build / and /how-to-use informational pages
- └─ Implement /kiosk/[sessionId] with near-black high-contrast palette, icon-redundant status coding, and resolve Open Item 6 (on-deck visibility)
- └─ Develop /live/[sessionId] mobile player view with QR code generator, warm palette, staged-status indicator
- └─ Build /admin/[sessionId] host console: three-zone layout (Courts, On-Deck, Queue), 2x2 selectable player-card grid, silent edit/replace actions, responsive single-column collapse

Phase 4: History, Hardening & Deployment (Weeks 7-8)

- └─ Build /history session archive, audit views, and CSV export pipeline (documenting the no-logging caveat per Section 9)
- └─ Conduct network degradation simulations (packet loss, offline recovery)
- └─ Resolve Open Item 8 (on-deck cap changes mid-session)
- └─ Run on-site pilot trials across 6 active pickleball courts, specifically validating olive-vs-terracotta contrast on physical TV hardware under venue lighting

13. Verification & Quality Acceptance Criteria

- **Court Scaling:** The system cleanly handles 1 to 6 courts without CSS grid distortion on 1080p and 4K displays.
- **On-Deck Cap Enforcement:** The number of on-deck slots never exceeds $\overline{\text{active_courts} - 1}$ (or the host override), and recomputes immediately when active court count

changes.

- **On-Deck Manual Call Only:** No code path auto-promotes an on-deck matchup to a court under any timing condition — this must be verified with an explicit test asserting that a court sitting `available` for an extended period does not self-populate from on-deck.
- **Player Card Selectability:** Every player within an active court or on-deck slot is independently tappable, opens the correct edit/replace action sheet, and updates without affecting other players in the same matchup.
- **No Edit Logging:** Verify no `INSERT` occurs against any audit/log table as a side effect of a player edit or replacement (guards against accidental reintroduction of logging).
- **Stopwatch Accuracy:** Court stopwatches and grace countdowns maintain temporal synchronization within _____ across host tablets, TV kiosks, and mobile phones using server timestamps (`started_at`, `summoned_at`).
- **Non-Alphabetical Sort:** Players with identical records must sort strictly by point differential, then volume, then check-in time, never by alphabetical name sorting.
- **Elo Precision:** Dynamic rating updates must reflect the logistic expectation curve with integer rounding and no floating-point drift.
- **Pair-Lock Integrity:** Players locked as a pair are never separated onto opposing teams or different courts.
- **Zero-Install Performance:** Unauthenticated players scanning the court QR code must load their personal queue status in under 3 seconds on standard 4G connections.
- **Scoring Mode Flexibility:** Toggling the scoring requirement to "Off" cleanly removes score entry modals from the host console and transitions courts immediately upon clicking "Complete Match."
- **Color Contrast:** Kiosk status colors (olive/terracotta/red against near-black) must meet WCAG AA contrast minimums, and every status must remain distinguishable with color removed (icon/shape only), verified via a grayscale render test.